

## 1 Prepare data

Kaggle: 20,257  
ISOT: 38,283

Normalize title and body  
Remove exact duplicates

Remove cross-data overlap  
Define training / test sets

Labels: real 0, fake 1



## 2 Select models

**Training data only**

Fit TF-IDF inside folds  
Unigrams and bigrams

Tune 13 classifiers  
Three-fold CV by MCC

Primary choice: LinearSVC



## 3 Test generalization

Random Kaggle split  
Known-author-disjoint split

Kaggle to ISOT  
ISOT to Kaggle

Author / title / body ablation  
Freeze protocol settings



## 4 Evaluate results

Accuracy, F1, MCC, AUC  
Confusion matrices and ROC

Random-test bootstrap  
McNemar with Holm

Separate all-Kaggle export  
For Django integration